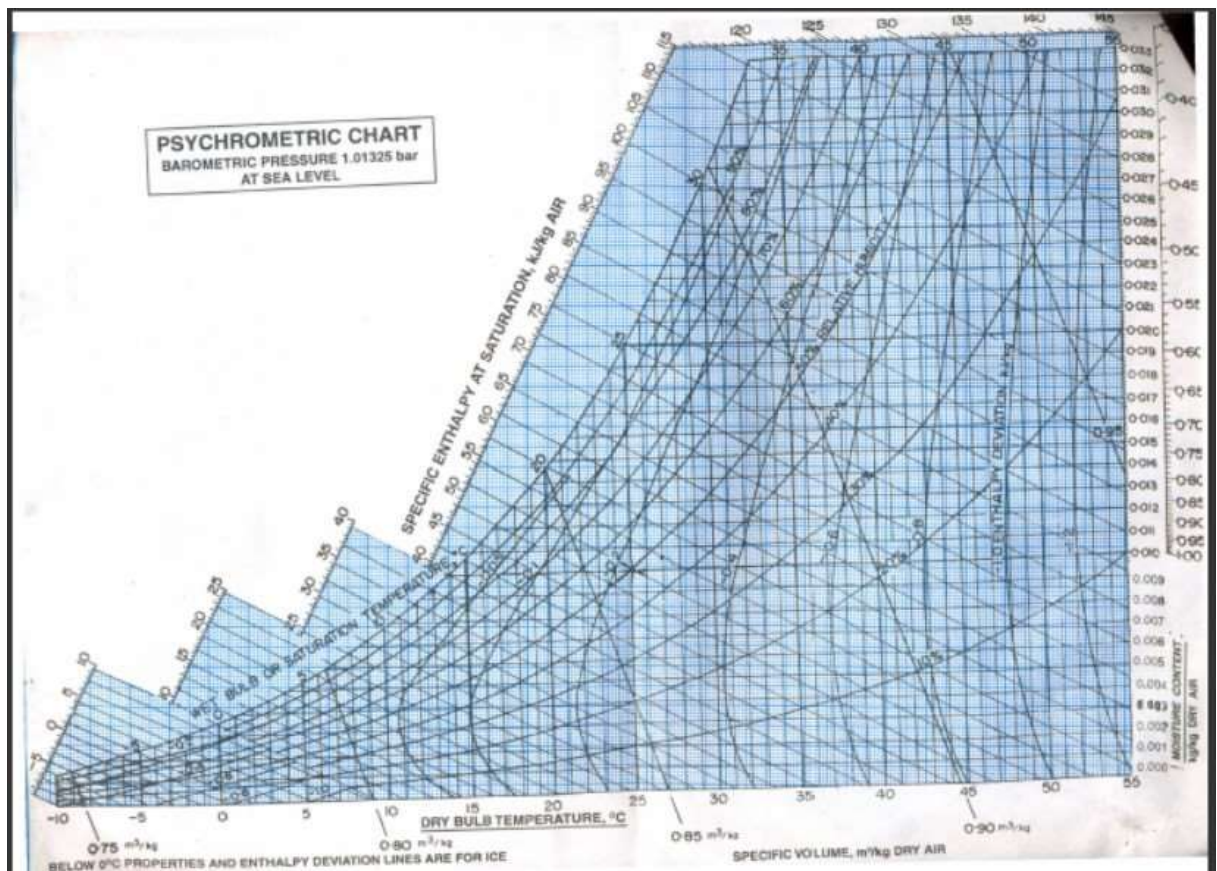


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Exam-2023 Course: B. Tech. Branch: Mechanical & allied Semester: VI Subject Code & Name: Applied Thermodynamics- II (BTME603_Y19) Max Marks: 60 Date: 20.01.2024 Duration: 2:00 pm to 05:00 pm					
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of steam tables, Refrigerant property tables, Psychrometric charts and non –non-programmable calculators is allowed					
				(Level/CO)	Marks
Q. 1	Solve Any Two of the following.				12
A)	Draw a valve timing diagram for a typical 4-stroke Petrol engine. Explain, what is valve overlap.			Understand/ CO1	6
B)	Compare 4-stroke and 2-stroke cycle engines.			Understand/ CO1	6
C)	Oil engine working on the Diesel cycle has a compression ratio of 14. The fuel cut-off takes place at 6 % of stroke. Calculate the cut-off ratio and diesel cycle efficiency. State the assumptions made, if any. $r_c=1.78$, $\eta_{TH}=60.43\%$			Apply /CO1	6
Q.2	Solve Any Two of the following.				12
A)	Draw the schematic diagram Diesel engine Power Plant and State various applications of the Diesel Engine Power Plant			Understand/ CO5	6
B)	Explain with a neat labelled diagram of a typical water-cooled system used in a four-cylinder engine.			Apply/ CO2	6
C)	Explain the catalytic converter used in IC engine. Discuss various types of alternative fuels used for SI and CI engines in India.			Understand/ CO2	6
Q. 3	Solve Any Two of the following.				12
A)	What is the difference between vapour absorption and vapour compression systems?			Apply/ CO3	6
B)	A vapour compression refrigerator uses R-40 and operates between temp. limits of -10°C and 45°C . At the entry to the compressor, the refrigerant is dry saturated and after compression, it acquires a temperature of 60°C . Find The C.O.P. of a refrigerator. The relevant properties of R-40 are as follows			Apply/ CO3	6
	Saturation Temp. in $^\circ\text{C}$	Enthalpy in KJ/Kg		Entropy in KJ/Kg K	
		Liquid	Vapour	Liquid	Vapour
	-10	45.4	460.7	0.183	1.637
	45	133	483.6	0.485	1.587

C)	Discuss the desirable properties of refrigerants used in vapour compression refrigeration systems.	Understand/ CO3	6
Q.4	Solve Any Two of the following.		12
A)	Assume that the outside air temperature is 32°C with a relative humidity $\phi = 60\%$. Use the psychrometric chart to determine the 1. Specific humidity 2. The enthalpy 3. The wet-bulb temperature 4. The dew-point temperature and 5. The specific volume of the dry air . 6. Indicate all the values determined on the chart.	Apply/CO4	6
B)	Explain following terms: 1) DBT 2) WBT 3) Specific humidity 4) Relative humidity	Remember/ CO4	6
C)	Explain the following source of energy for the power plant i) Fossil fuels, ii) Petroleum products, iii) Tidal	Understand/ CO5	6
Q. 5	Solve Any Two of the following.		12
A)	Write short notes on 1. Topping cycle 2. Rankine cycle	Remember/ CO5	6
B)	Explain with a schematic diagram of a typical thermal power plant.	Apply/CO6	6
C)	Draw a schematic diagram of a typical nuclear power plant. Write down the guide-lines for the selection of a site for this type of power plant.	Remember/ CO6	6
	*** End ***		

The grid and the borders of the table will be hidden before final printing.



[Note:- Use the above chart for Q.No. 4 (A)]